

MARKING SCHEME

NETWORKING CAT 3 MOD 3

QUESTION ONE (20 MARKS)

1. Differentiate between the OSI Model and the TCP/IP Model. (2 Marks)

Answer	Marks
OSI model has 7 layers while TCP/IP model has 4 layers	1
OSI is a reference model while TCP/IP is a practical protocol suite used on the Internet	1

Total: 2 Marks

2. State any two functions of the Network Layer in the OSI model. (2 Marks)

Any two:

- Routing packets between networks
- Logical addressing (IP addressing)
- Path determination
- Packet forwarding

(1 Mark each)

Total: 2 Marks

3. Define the following terms. (2 Marks)

a) IPv4 Address (1 Mark)

A 32-bit logical address used to identify devices on a network.

b) IPv6 Address (1 Mark)

A 128-bit logical address designed to replace IPv4 and provide a larger address space.

Total: 2 Marks

4. Identify the class of each IP address. (2 Marks)

Address	Class	Marks
10.0.0.1	Class A	1
172.16.5.10	Class B	1

Total: 2 Marks

5. Differentiate between Static IP Address and Dynamic IP Address. (2 Marks)

Answer	Marks
Static IP is manually assigned and remains unchanged	1
Dynamic IP is automatically assigned by DHCP and may change over time	1

Total: 2 Marks

6. State any three advantages of IPv6 over IPv4. (3 Marks)

Any three:

- Larger address space
- Improved security through IPsec support
- Better routing efficiency
- Auto-configuration capability
- Reduced need for NAT

(1 Mark each)

Total: 3 Marks

7. Explain the term CIDR (Classless Inter-Domain Routing). (2 Marks)

CIDR is a method of IP addressing that allows flexible allocation of IP addresses using subnet masks rather than fixed address classes.

- Correct explanation = 2 Marks

Total: 2 Marks

8. State any two advantages of subnetting. (2 Marks)

Any two:

- Efficient utilization of IP addresses
- Reduced broadcast traffic
- Improved network performance
- Enhanced security and management

(1 Mark each)

Total: 2 Marks

9. Differentiate between a Switch and a Router. (3 Marks)

Answer	Marks
Switch connects devices within the same network/LAN	1
Router connects different networks	1
Switch uses MAC addresses while Router uses IP addresses	1

Total: 3 Marks

QUESTION TWO (15 MARKS)

10. Outline three steps involved in the initial setup and configuration of a Cisco router. (3 Marks)

Any three:

- Connect to the router using console cable

- Enter privileged EXEC mode
- Enter global configuration mode
- Configure hostname
- Configure interfaces and IP addresses
- Save configuration

(1 Mark each)

Total: 3 Marks

11. Write the Cisco IOS commands used to: (3 Marks)

a) Enter global configuration mode (1 Mark)

`configure terminal`

b) Assign an IP address to an interface (1 Mark)

`ip address 192.168.1.1 255.255.255.0`

c) Save the running configuration (1 Mark)

`copy running-config startup-config`

or

`write memory`

Total: 3 Marks

12. Explain the purpose of the following routing protocols. (3 Marks)

a) RIP (1 Mark)

Uses hop count to determine the best route.

b) OSPF (1 Mark)

Uses link-state information to determine shortest paths.

c) EIGRP (1 Mark)

Cisco proprietary protocol that provides fast convergence and efficient routing.

Total: 3 Marks

13. State any two benefits of implementing VLANs in a network. (2 Marks)

Any two:

- Improved security
- Reduced broadcast traffic
- Easier network management
- Better performance

(1 Mark each)

Total: 2 Marks

14. What is an Access Control List (ACL) and why is it used in network security? (2 Marks)

Answer	Marks
ACL is a set of rules used to permit or deny network traffic	1
Used to enhance security and control access to resources	1

Total: 2 Marks

15. Differentiate between NAT and PAT. (2 Marks)

Answer	Marks
NAT translates one private IP address to one public IP address	1
PAT translates multiple private IP addresses to a single public IP using port numbers	1

Total: 2 Marks

QUESTION THREE (15 MARKS)

16. State any two benefits of implementing Port Security on a switch. (2 Marks)

Any two:

- Prevents unauthorized device connections
- Restricts access based on MAC addresses
- Enhances network security
- Reduces network attacks

(1 Mark each)

Total: 2 Marks

17. Explain the purpose of Spanning Tree Protocol (STP) in a switched network. (2 Marks)

- Prevents switching loops in a network (1 Mark)
- Ensures a loop-free topology by blocking redundant paths (1 Mark)

Total: 2 Marks

18. What is Link Aggregation Control Protocol (LACP) and why is it used? (2 Marks)

Answer	Marks
LACP combines multiple physical links into a single logical link	1
Used to increase bandwidth and provide redundancy	1

Total: 2 Marks

19. Explain any three configurations performed when setting up a Wireless Access Point (AP). (3 Marks)

Any three:

- Configure SSID
- Configure security settings (WPA2/WPA3)
- Assign IP address
- Select wireless channel
- Configure DHCP settings

- Set administrator password

(1 Mark each)

Total: 3 Marks

20. State the functions of the following network services. (2 Marks)

a) DHCP (1 Mark)

Automatically assigns IP addresses and network settings to devices.

b) DNS (1 Mark)

Translates domain names into IP addresses.

Total: 2 Marks

21. What is an SSID and why is it important in wireless networking? (2 Marks)

Answer	Marks
SSID is the name/identifier of a wireless network	1
Allows users to identify and connect to the correct wireless network	1

Total: 2 Marks

22. Describe two measures that can be used to secure a wireless network. (2 Marks)

Any two:

- Use WPA2/WPA3 encryption
- Use strong passwords
- Change default credentials
- Disable WPS
- Enable MAC filtering

(1 Mark each)

Total: 2 Marks

23. Define the term Network Privileges and state one example. (2 Marks)

Answer	Marks
Network privileges are permissions granted to users to access network resources	1
Example: Administrator privileges, Read-only access, File sharing access	1

Total: 2 Marks

QUESTION FOUR: WIRESHARK AND NETWORK SECURITY (10 MARKS)

24. Using Wireshark packet captures, explain the TCP Three-Way Handshake. (4 Marks)

a) SYN (1 Mark)

Client sends SYN packet to initiate a connection.

b) SYN-ACK (1 Mark)

Server responds with SYN-ACK acknowledging the request.

c) ACK (1 Mark)

Client sends ACK confirming receipt and completing the connection setup.

d) Purpose of TCP Three-Way Handshake (1 Mark)

Establishes a reliable connection before data transmission begins.

Total: 4 Marks

25. What is Wireshark and state two uses of Wireshark in network administration? (3 Marks)

Definition (1 Mark)

Wireshark is a packet analyzer used to capture and analyze network traffic.

Uses (Any Two × 1 Mark)

- Network troubleshooting
- Monitoring network traffic
- Security analysis
- Protocol analysis
- Detecting network issues

Total: 3 Marks

26. A network administrator wants to display only TCP packets in Wireshark. (3 Marks)

a) Display filter (1 Mark)

tcp

b) One advantage of packet analysis during network troubleshooting (2 Marks)

Any one well-explained advantage:

- Helps identify network faults
- Detects packet loss
- Diagnoses protocol errors
- Identifies security threats
- Improves network performance analysis

(2 Marks)

Total: 3 Marks